Uploading the Source Code

For this project, we have two different sketches—one to simply display the time from the RTC in a text-based, easily readable format and a second to let us set the time in the device using the Serial Monitor. The code might seem a little overwhelming at first, but the first sketch is written to be modular, allowing readers to use it for whatever nefarious means they might dream up. The second sketch is more of a utility than anything, although seeing how to set the time and date from within the code could be handy for something I haven't considered. The format for the text to be displayed should look like the following:

```
Thursday July 7, 2011 8:35 PM
```

While the RTC is a 24-hour clock by default, we have used a little `if` statement to convert this to 12-hour format rather than trying to set the 12-hour clock in the device. To read the date and time, we need to read 7 bytes of data in a row corresponding to a specific order of units, such as second, hour, minute, and so on. The data is inconveniently sent in Binary Coded Decimal or BCD format. This means that each digit is sent as a 4-bit binary number so a decimal `1` would be sent as `0001` and `5` would be sent as `0101`. We don't have to worry too much about this as we have a nifty little function to convert this for us. The rest of the code consists of fairly basic `print()` functions so that we can see the data on our serial monitor.

Following in Listings 10-3 and 10-4 are the two sketches for this project. We can start with the simpler sketch to see how to read the data from the real-time clock and see what kind of data is spit back out. When we have a sense for this code, we will move on to the sketch in Listing 10-4 that really only needs to be run once to set the time for your location. This will demonstrate how to send the time to the device and all of the code that goes along with it. Listing 10-3 will concentrate the discussion on the first sketch while pointing out significant differences in the second.

Listing 10-3. *Serial Time Clock Display Source Code*

```
#include <Wire.h>

const int DS1307 = 0x68;

const char* days[] =
  {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"};
const char* months[] =
  {"January", "February", "March", "April", "May", "June", "July", "August",
   "September", "October", "November", "December"};

byte second = 0;
byte minute = 0;
byte hour = 0;
byte weekday = 0;
byte monthday = 0;
byte month = 0;
byte year = 0;

byte lastMinute = 0;
```